WORKED SOLUTION

STEP I 2013, Question 6

Pure • Binomial coefficients

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q6

The key idea

Pascal's rule is the whole engine. Each B_n is a sum of binomial coefficients, and if we split every term of B_{n+2} with Pascal's rule, the pieces regroup into exactly the terms of B_{n+1} and B_n. The only care needed is at the ends of the sums, where Pascal's rule doesn't apply and a lone 1 has to be matched up by hand. Because B_n is defined differently for even and odd n, we do the two cases separately.

Solution

Pascal's rule

Since $(1+x)^{n+1} = (1+x)(1+x)^n$, the coefficients of x^r on the two sides agree. On the left the coefficient is $\binom{n+1}{r}$. On the right,

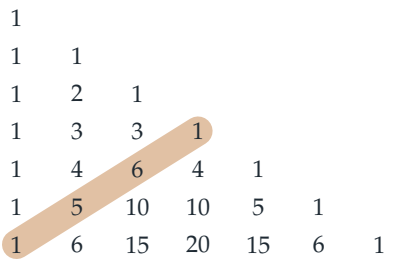
$$(1+x)\sum_{s=0}^{n}\binom{n}{s}x^s = \sum_{s=0}^{n}\binom{n}{s}x^s + \sum_{s=0}^{n}\binom{n}{s}x^{s+1},$$

and for $1 \le r \le n$ the first sum contributes $\binom{n}{r}$ (from $s=r$) and the second contributes $\binom{n}{r-1}$ (from $s=r-1$). Hence

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r} \qquad (1 \le r \le n).$$

The recurrence

It helps to see what B_n is: it adds up a "shallow diagonal" of Pascal's triangle, $\binom{n}{0} + \binom{n-1}{1} + \binom{n-2}{2} + \cdots$, stopping at the last term whose top is at least its bottom. With the triangle left-justified, the terms making up $B_6 = 1+5+6+1 = 13$ lie on a straight line:



(The amber band picks out $\binom{6}{0}, \binom{5}{1}, \binom{4}{2}, \binom{3}{3}$.)

Even case, $n = 2m$ ($m \geq 0$). We need $B_{2m+2} = B_{2m+1} + B_{2m}$. By definition

$$B_{2m+2} = \sum_{j=0}^{m+1} \binom{2m+2-j}{j}.$$

The end terms are $j = 0$, giving $\binom{2m+2}{0} = 1$, and $j = m+1$, giving $\binom{m+1}{m+1} = 1$. For $1 \leq j \leq m$ we use Pascal's rule with top $2m+1-j$ and bottom j; this is allowed because $j \leq m$ gives $j \leq 2m+1-j$. So

$$B_{2m+2} = \underbrace{1 + \sum_{j=1}^{m} \binom{2m+1-j}{j}}_{\text{(a)}} + \underbrace{\sum_{j=1}^{m} \binom{2m+1-j}{j-1} + 1}_{\text{(b)}}.$$

In (a), the 1 is $\binom{2m+1}{0}$, the missing $k = 0$ term, so (a) is $\sum_{k=0}^{m} \binom{2m+1-k}{k} = B_{2m+1}$.

> **Ask yourself** What should the second sum become, and where does the spare 1 go?

Put $i = j - 1$ in (b): the sum becomes $\sum_{i=0}^{m-1} \binom{2m-i}{i}$, which is B_{2m} with its last term $i = m$ missing. That last term is $\binom{m}{m} = 1$, exactly the spare 1. So (b) $= B_{2m}$, and $B_{2m+2} = B_{2m+1} + B_{2m}$. (When $m = 0$ the sums are empty and this reads $2 = 1 + 1$.)

Odd case, $n = 2m+1$ ($m \geq 0$). We need $B_{2m+3} = B_{2m+2} + B_{2m+1}$. Now

$$B_{2m+3} = \sum_{k=0}^{m+1} \binom{2m+3-k}{k}.$$

Only the $k = 0$ term, $1 = \binom{2m+2}{0}$, is left alone. For $1 \leq k \leq m+1$, Pascal's rule with top $2m+2-k$ and bottom k is allowed, since $k \leq m+1$ gives $k \leq 2m+2-k$. Therefore

$$\begin{aligned} B_{2m+3} &= \binom{2m+2}{0} + \sum_{k=1}^{m+1} \binom{2m+2-k}{k} + \sum_{k=1}^{m+1} \binom{2m+2-k}{k-1} \\ &= \sum_{j=0}^{m+1} \binom{2m+2-j}{j} + \sum_{i=0}^{m} \binom{2m+1-i}{i} \qquad (i = k-1) \\ &= B_{2m+2} + B_{2m+1}. \end{aligned}$$

This time no end terms are left over.

Every $n \geq 0$ is of the form $2m$ or $2m+1$, so $B_{n+2} - B_{n+1} = B_n$ for all $n = 0, 1, 2, \ldots$, as required.

Relation with the Fibonacci numbers

Directly, $B_0 = \binom{0}{0} = 1$ and $B_1 = \binom{1}{0} = 1$. The Fibonacci numbers start $F_1 = 1$, $F_2 = 1$. So $B_0 = F_1$ and $B_1 = F_2$, and both sequences obey the same recurrence "each term is the sum of the previous two". By induction (if $B_n = F_{n+1}$ and $B_{n+1} = F_{n+2}$ then $B_{n+2} = F_{n+2} + F_{n+1} = F_{n+3}$),

$$B_n = F_{n+1} \qquad \text{for all } n \geq 0.$$

The B sequence is the Fibonacci sequence shifted by one place: $1, 1, 2, 3, 5, 8, 13, \ldots$.

Checks

- Compute a few terms straight from the definition: $B_2 = 1 + 1 = 2$, $B_3 = 1 + 2 = 3$, $B_4 = 1 + 3 + 1 = 5$, $B_5 = 1 + 4 + 3 = 8$. These are $F_3, \ldots, F_6$.
- A larger one: $B_9 = \binom{9}{0} + \binom{8}{1} + \binom{7}{2} + \binom{6}{3} + \binom{5}{4} = 1 + 8 + 21 + 20 + 5 = 55 = F_{10}$.
- Pascal's rule spot check: $\binom{5}{2} + \binom{5}{1} = 10 + 5 = 15 = \binom{6}{2}$.

Common slips

- Applying Pascal's rule to the end terms such as $\binom{m+1}{m+1}$, which would produce $\binom{m}{m+1}$, outside the range $1 \leq r \leq n$ for which the rule was proved. Peel off the end terms first.
- Treating only one parity. The definitions of B_{2m} and B_{2m+1} differ, and the recurrence links an even-indexed term to odd-indexed ones, so both cases are needed.
- Getting the index shift wrong: the answer is $B_n = F_{n+1}$, not $B_n = F_n$ (check $B_0 = 1 \neq F_0$).

Beyond STEP

The identity $F_{n+1} = \sum_k \binom{n-k}{k}$ has a neat counting proof: both sides count the ways of tiling a $1 \times n$ strip with squares and dominoes (k is the number of dominoes, and choosing their positions among the $n - k$ tiles gives $\binom{n-k}{k}$). Proving an identity by counting the same set in two ways is a core technique in undergraduate combinatorics, and so is the generating-function view, in which $\sum F_n x^n = x/(1 - x - x^2)$. For more, try Wilf, *generatingfunctionology*, or Siklos, *Advanced Problems in Mathematics*.

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